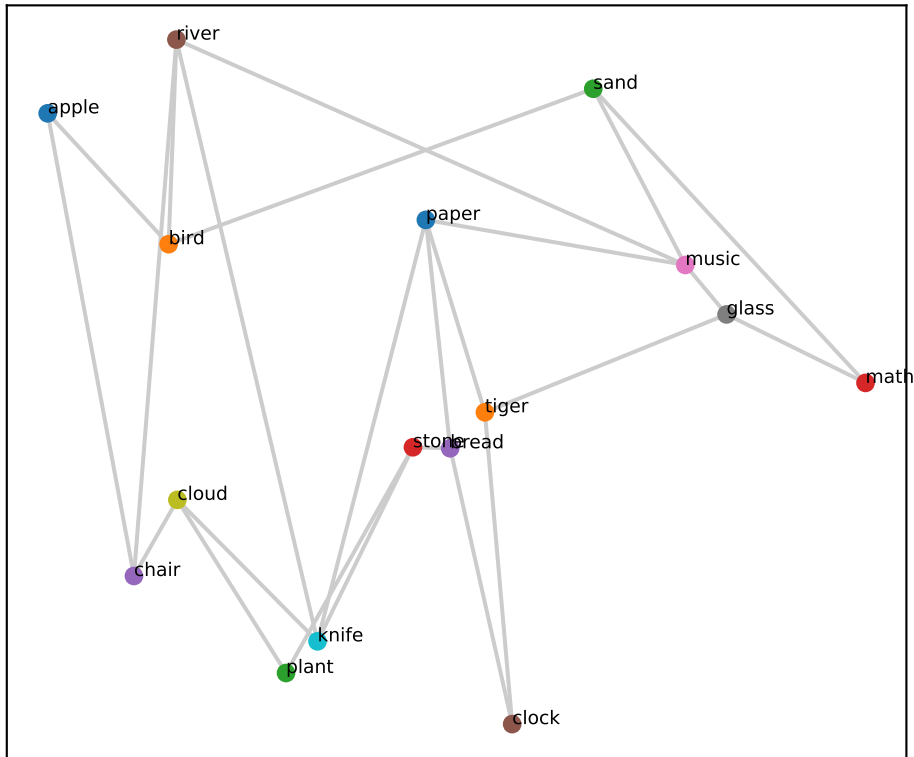
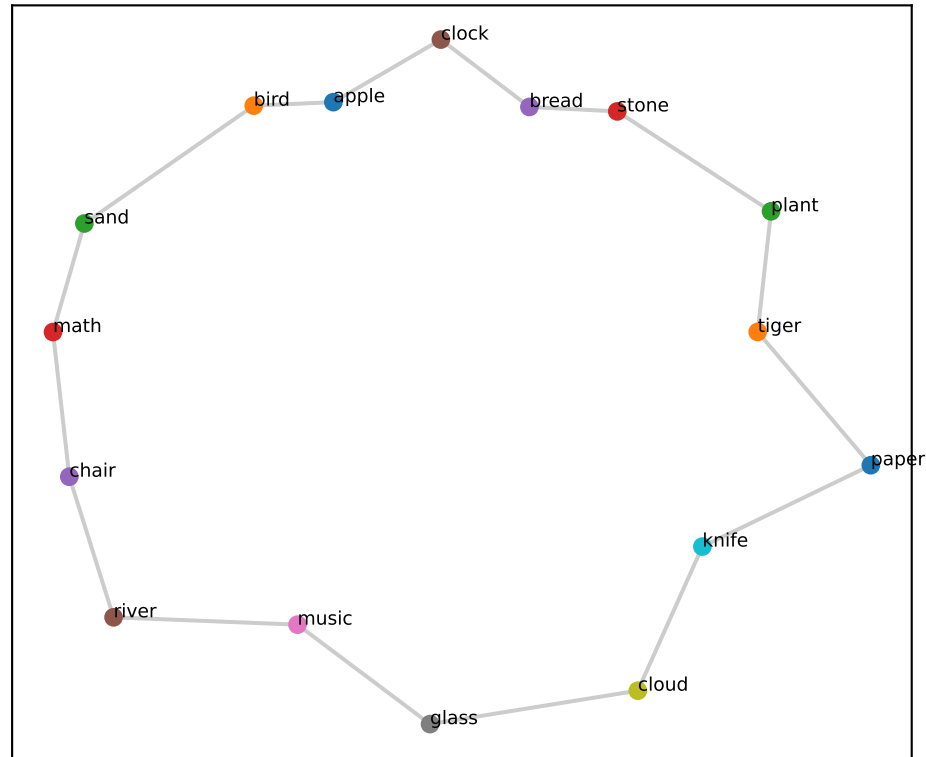


Llama Top: PCA top-2 (variance) | Bottom: best-2D (top-6 PCs -> coords); null = label-permutation through the fit

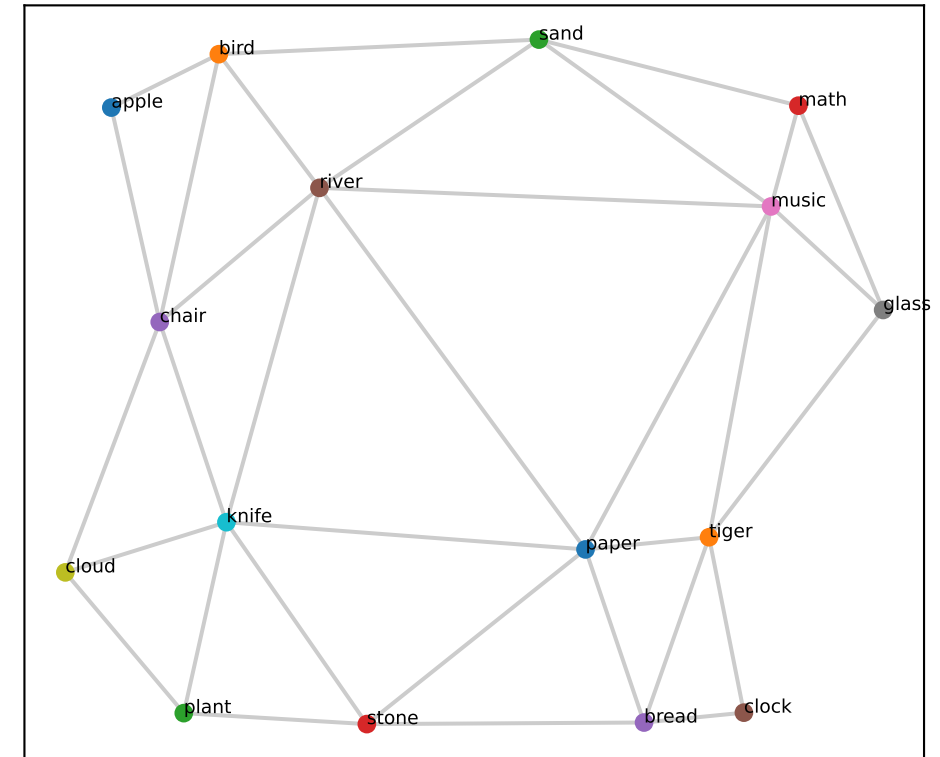
square_grid PCA-2D (RSA 0.48)



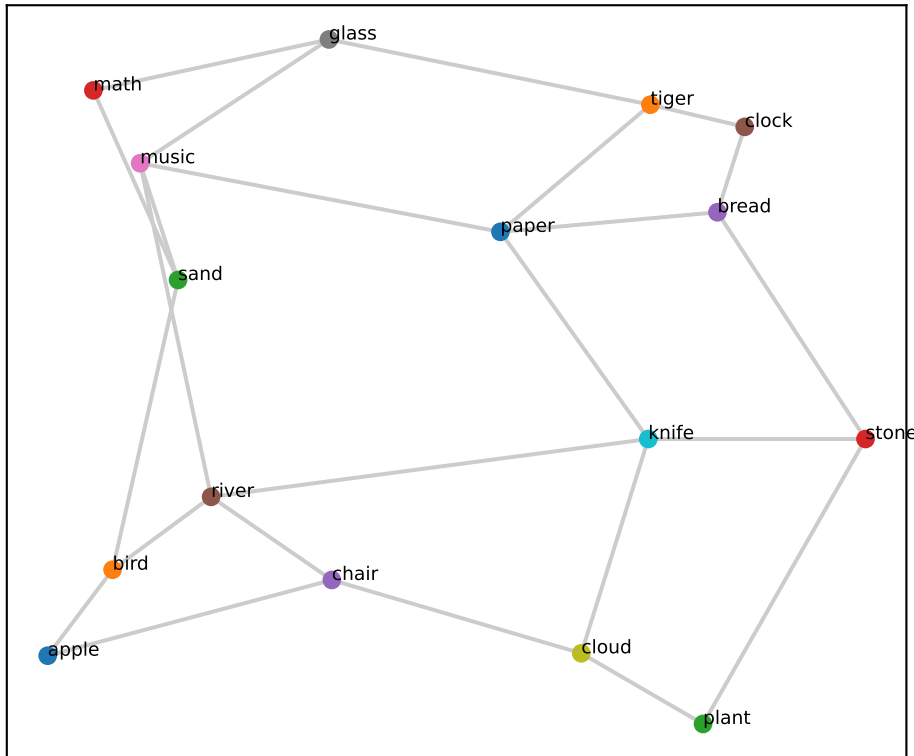
ring PCA-2D (RSA 0.92)



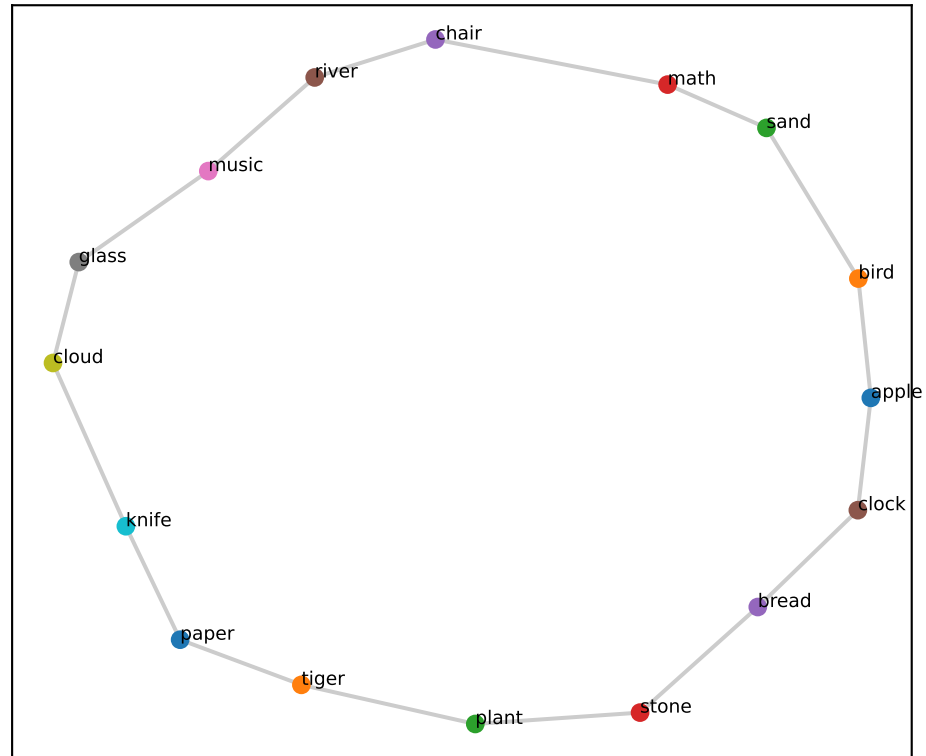
hex PCA-2D (RSA 0.86)



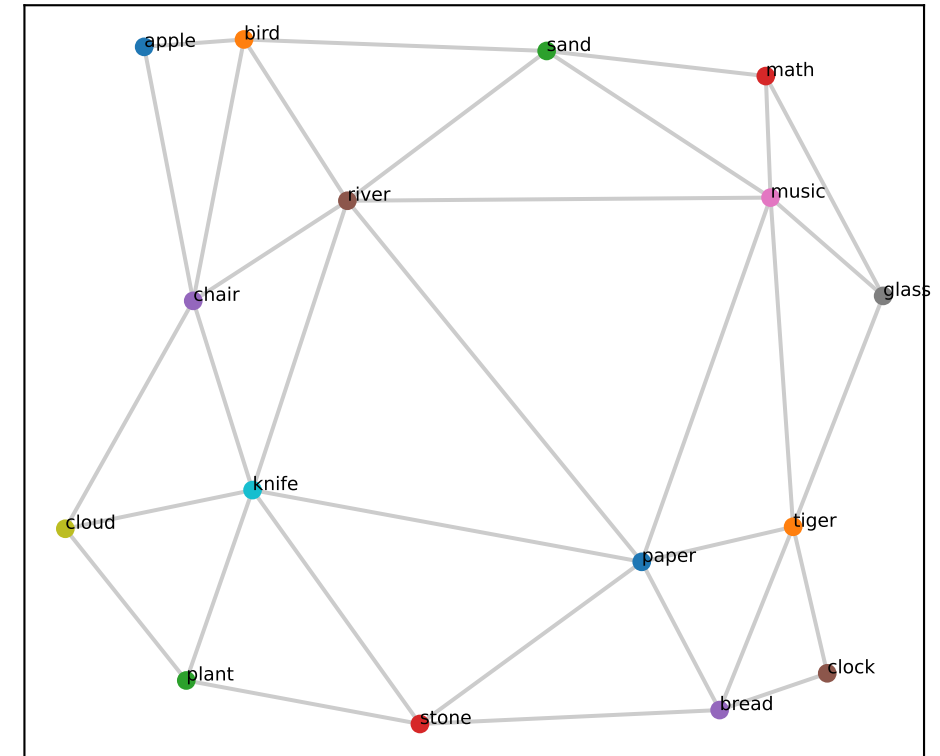
square_grid best-2D (RSA 0.81 | null95 0.48, $p < .001$)



ring best-2D (RSA 0.98 | null95 0.51, $p < .001$)

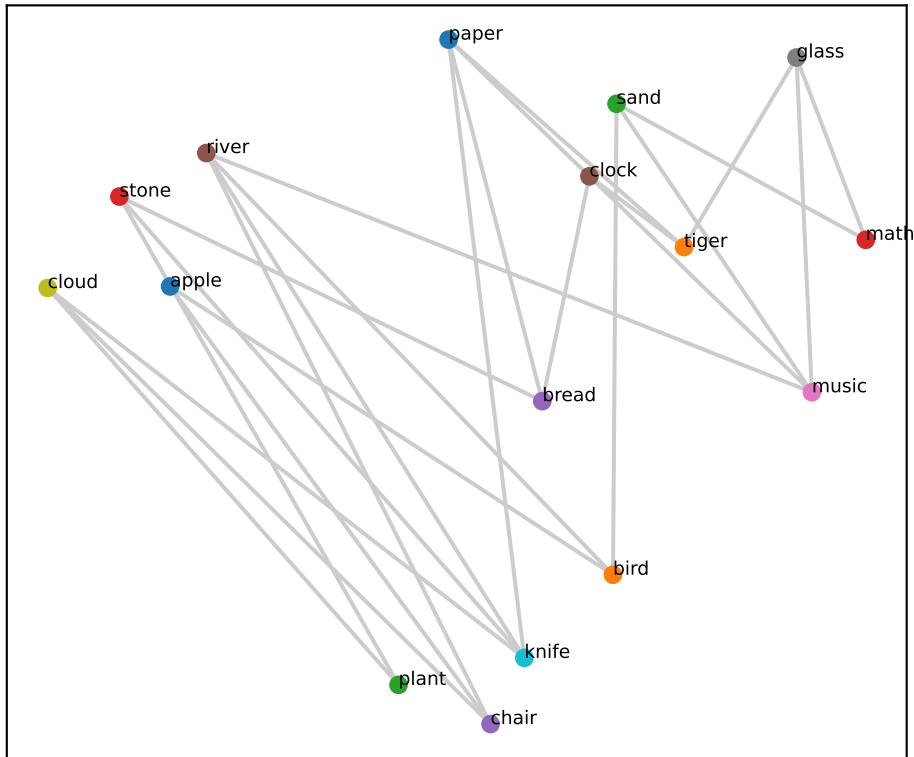


hex best-2D (RSA 0.88 | null95 0.46, $p < .001$)

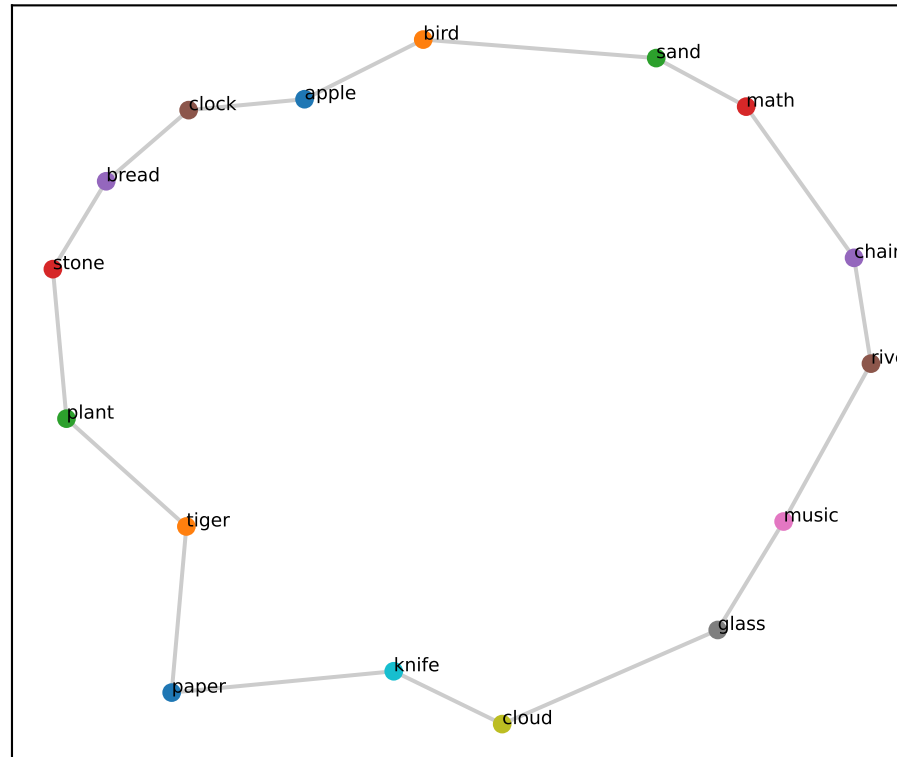


Gemma Top: PCA top-2 (variance) | Bottom: best-2D (top-6 PCs -> coords); null = label-permutation through the fit

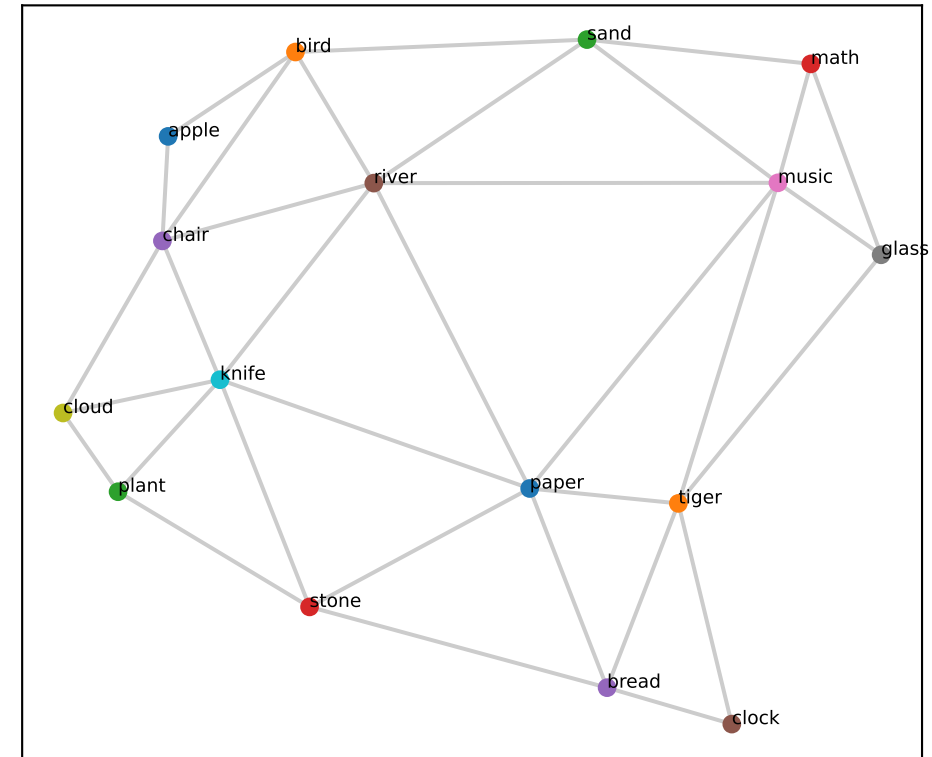
square_grid PCA-2D (RSA 0.24)



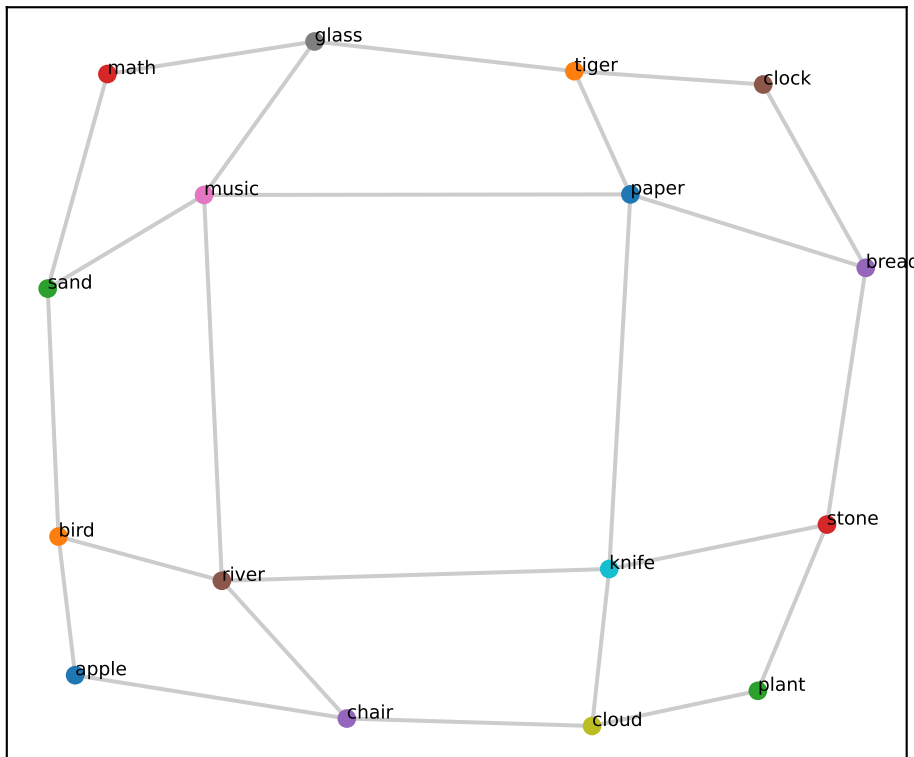
ring PCA-2D (RSA 0.94)



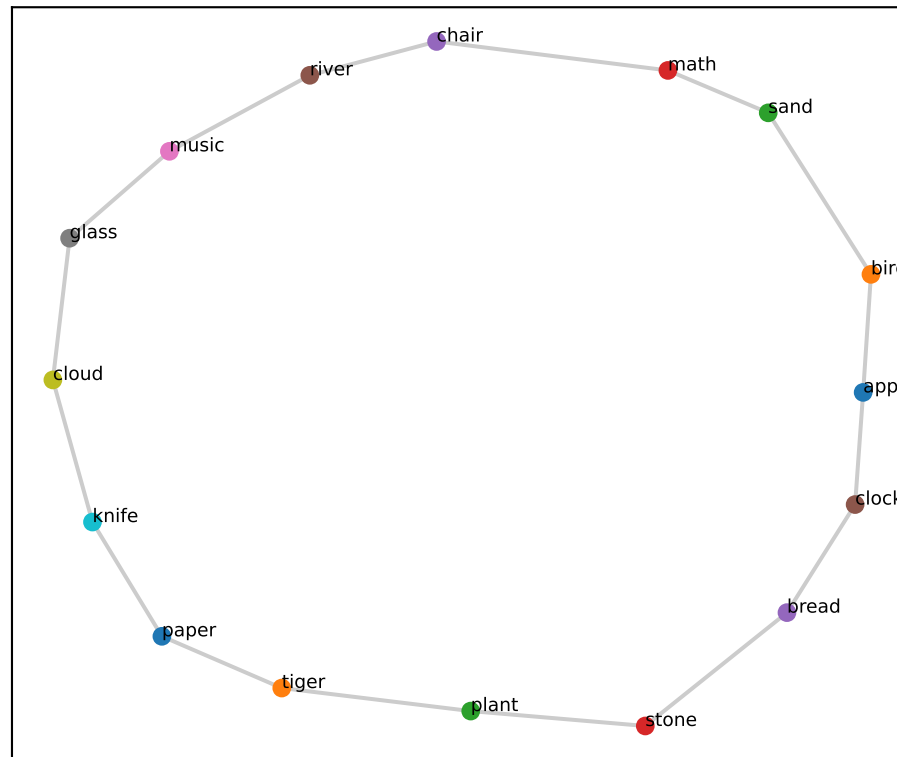
hex PCA-2D (RSA 0.87)



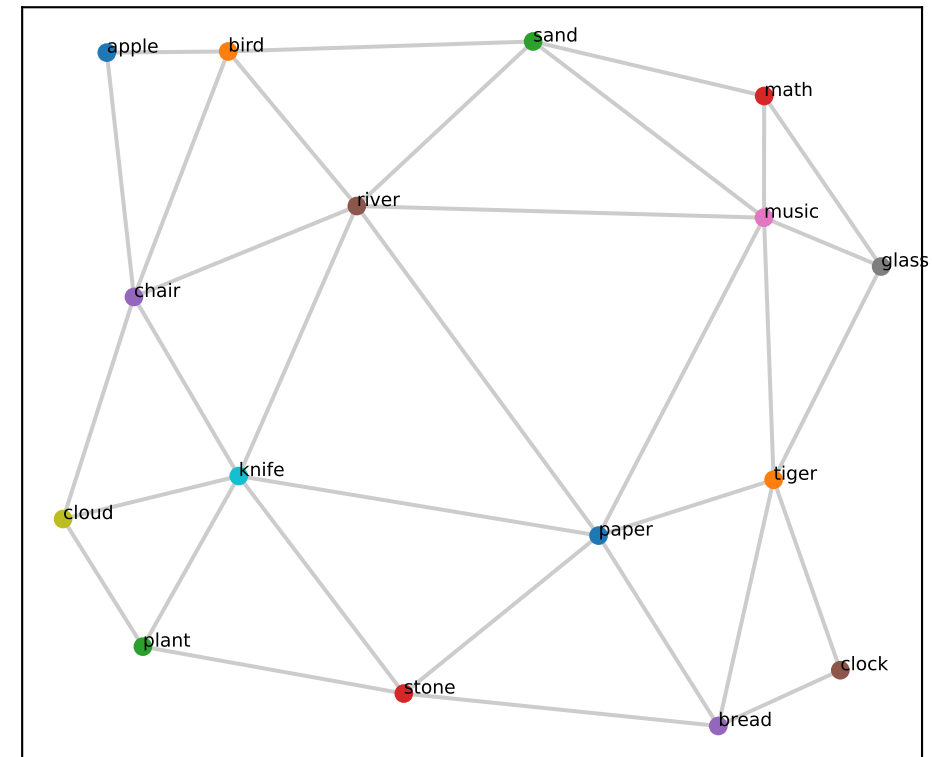
square_grid best-2D (RSA 0.88 | null95 0.47, $p < .001$)



ring best-2D (RSA 0.98 | null95 0.49, $p < .001$)

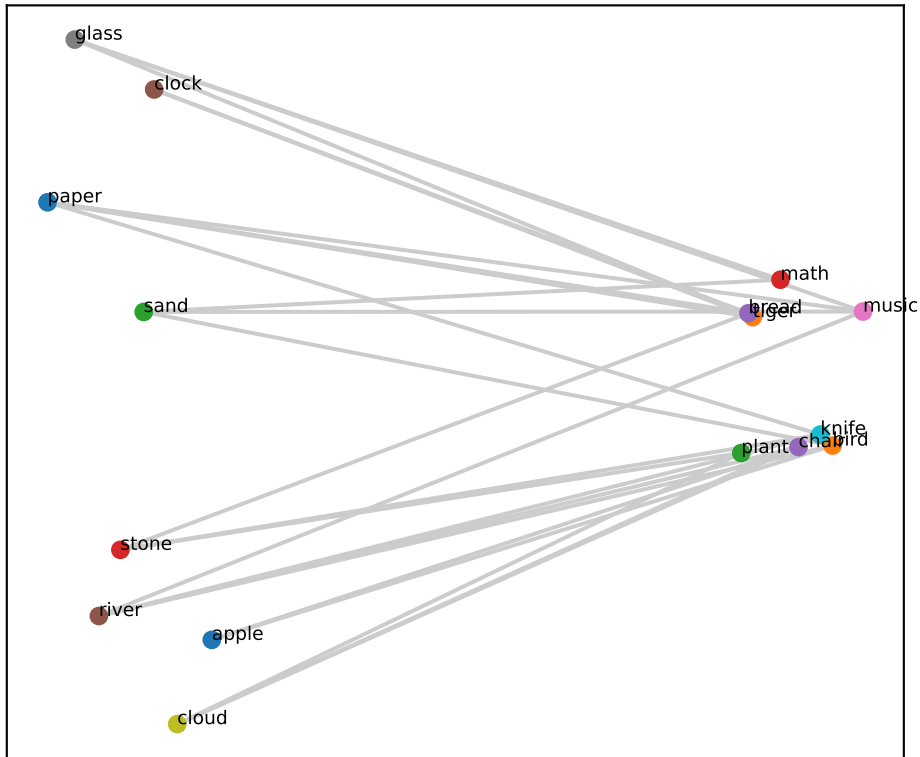


hex best-2D (RSA 0.90 | null95 0.47, $p < .001$)

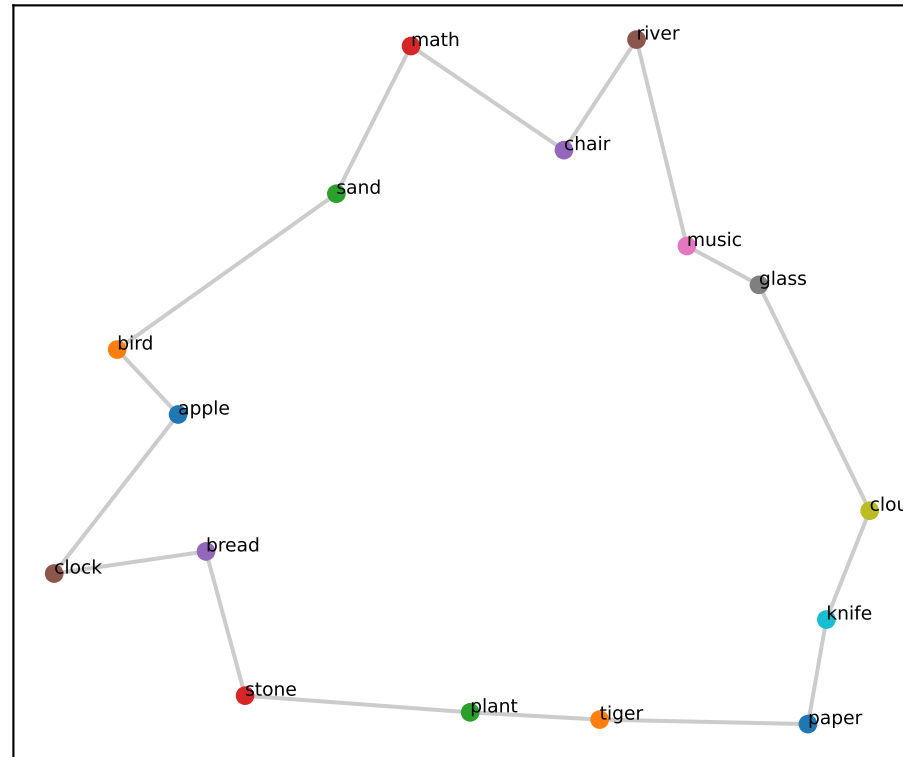


Qwen Top: PCA top-2 (variance) | Bottom: best-2D (top-6 PCs -> coords); null = label-permutation through the fit

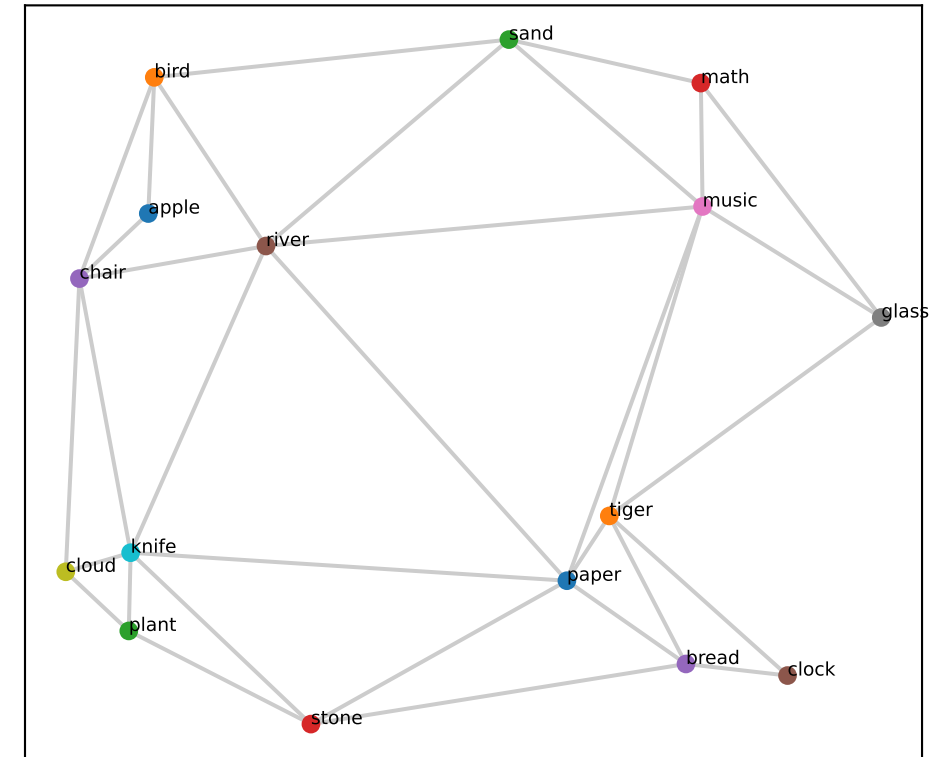
square_grid PCA-2D (RSA 0.06)



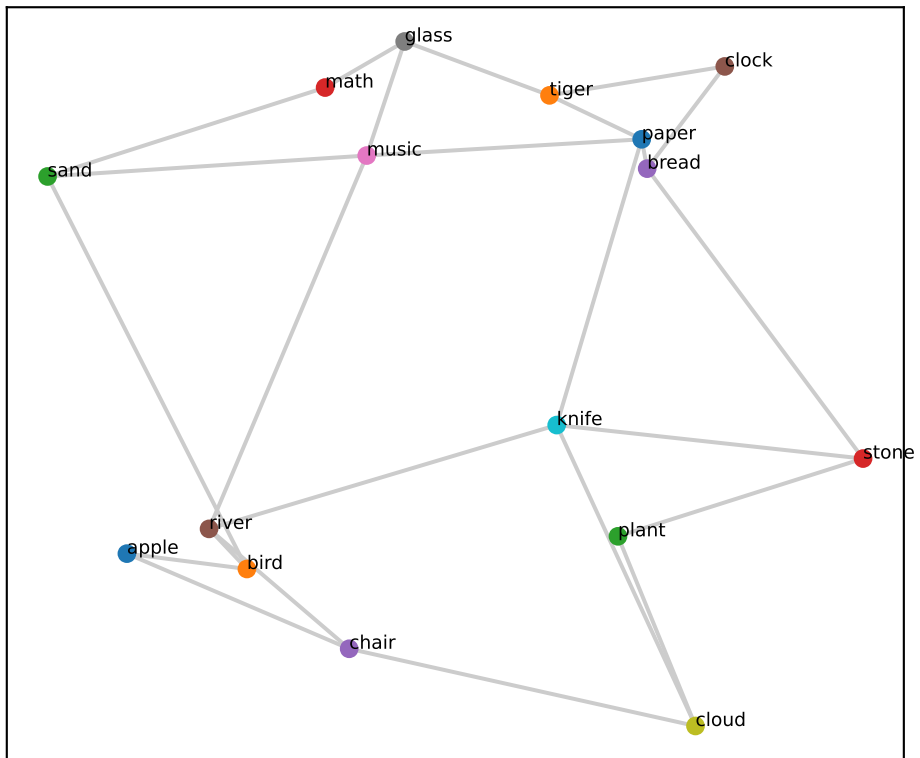
ring PCA-2D (RSA 0.89)



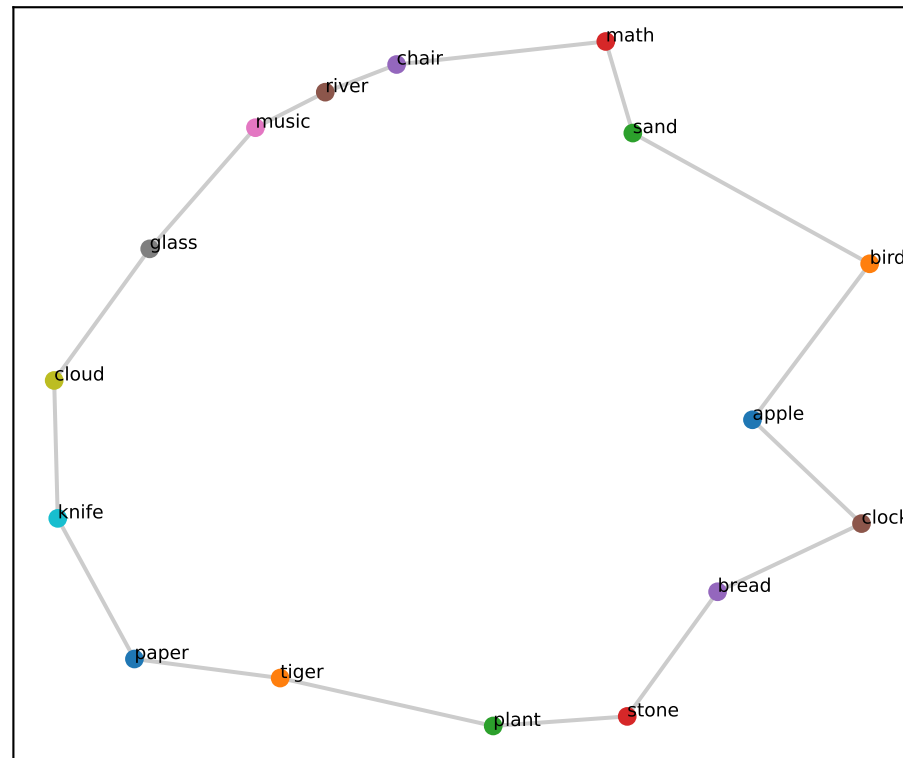
hex PCA-2D (RSA 0.82)



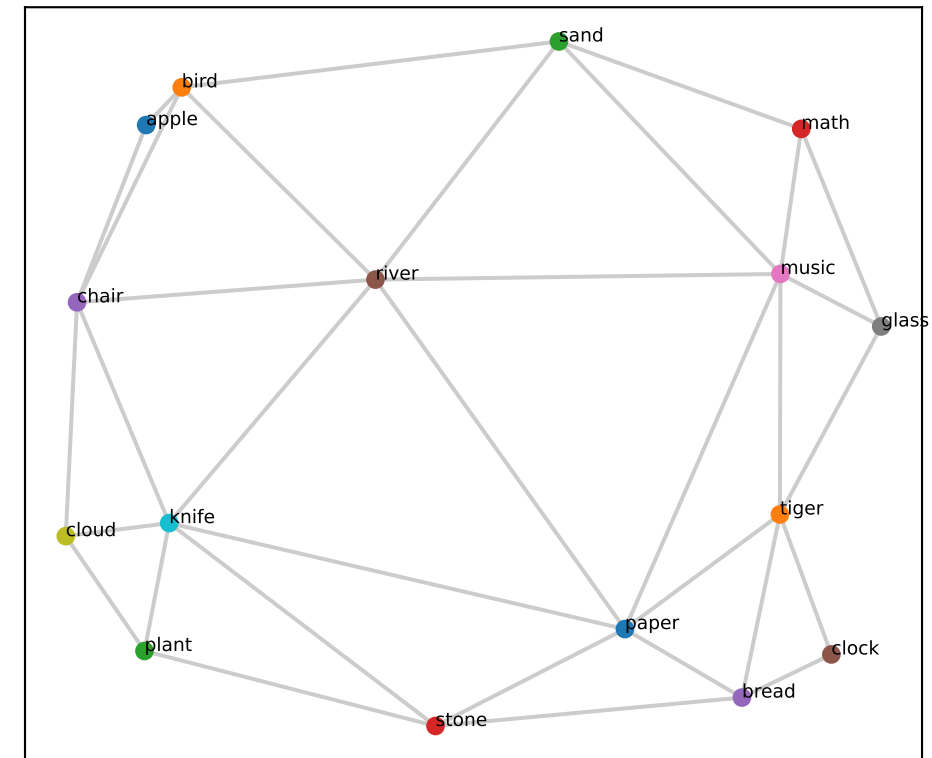
square_grid best-2D (RSA 0.72 | null95 0.45, p<.001)



ring best-2D (RSA 0.94 | null95 0.49, p<.001)

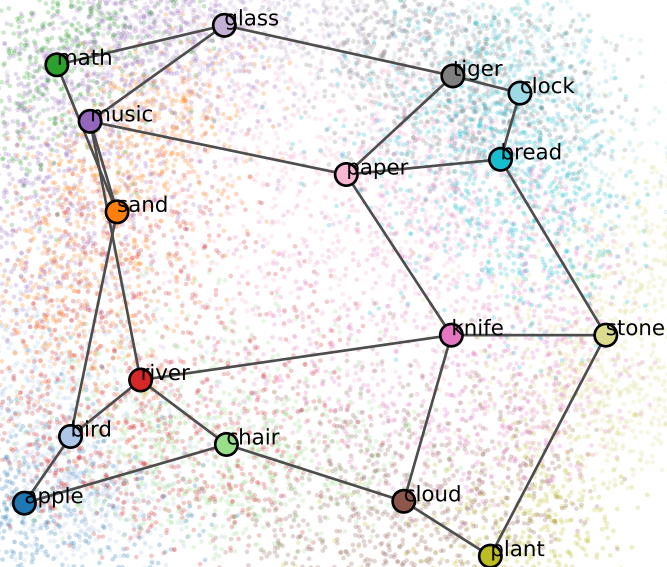


hex best-2D (RSA 0.85 | null95 0.49, p<.001)

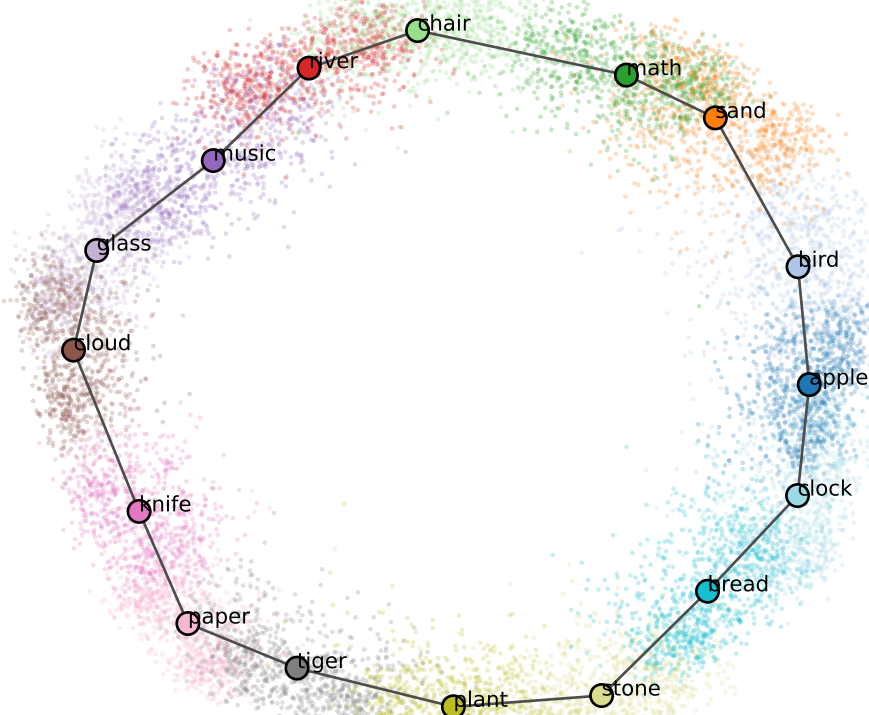


Llama — every occurrence on the two best-2D axes (dots=occurrences, markers=node means)

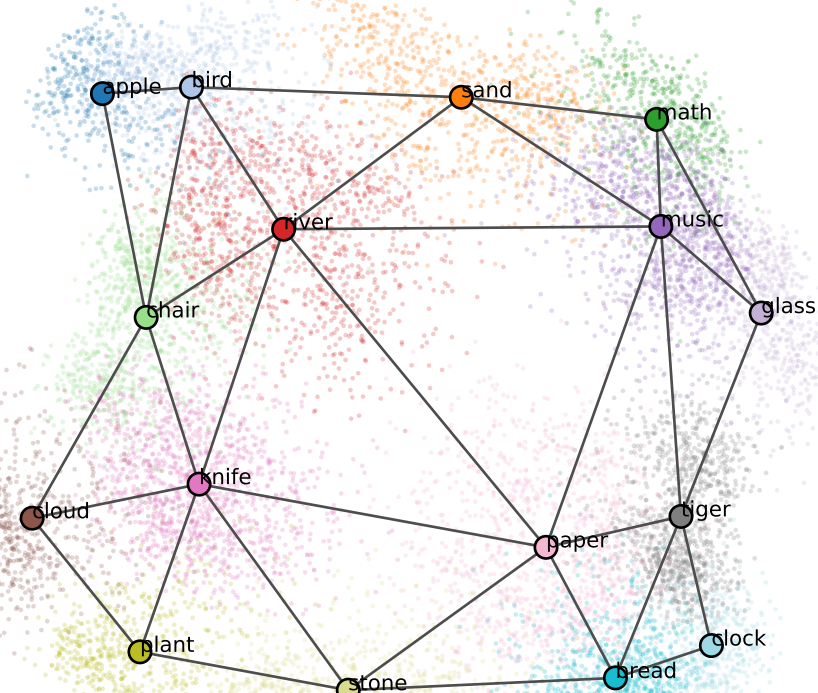
square_grid best-2D (RSA 0.81, $p < .001$)



ring best-2D (RSA 0.98, $p < .001$)

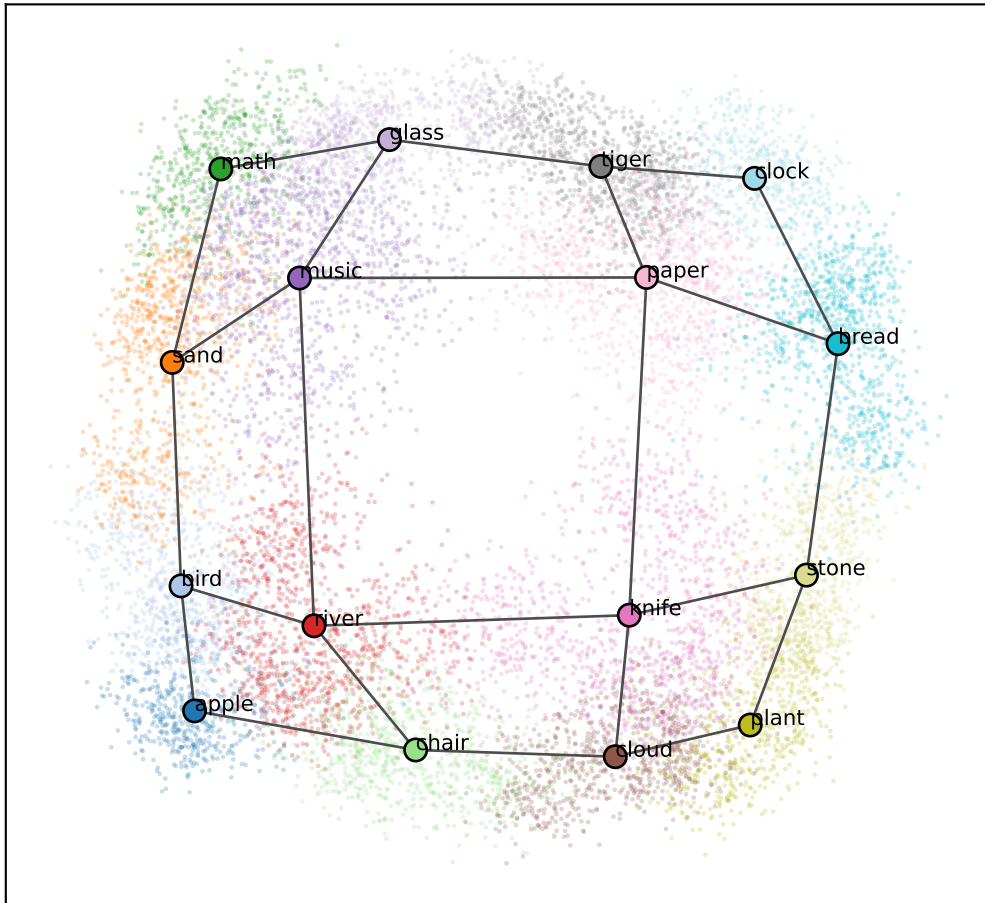


hex best-2D (RSA 0.88, $p < .001$)

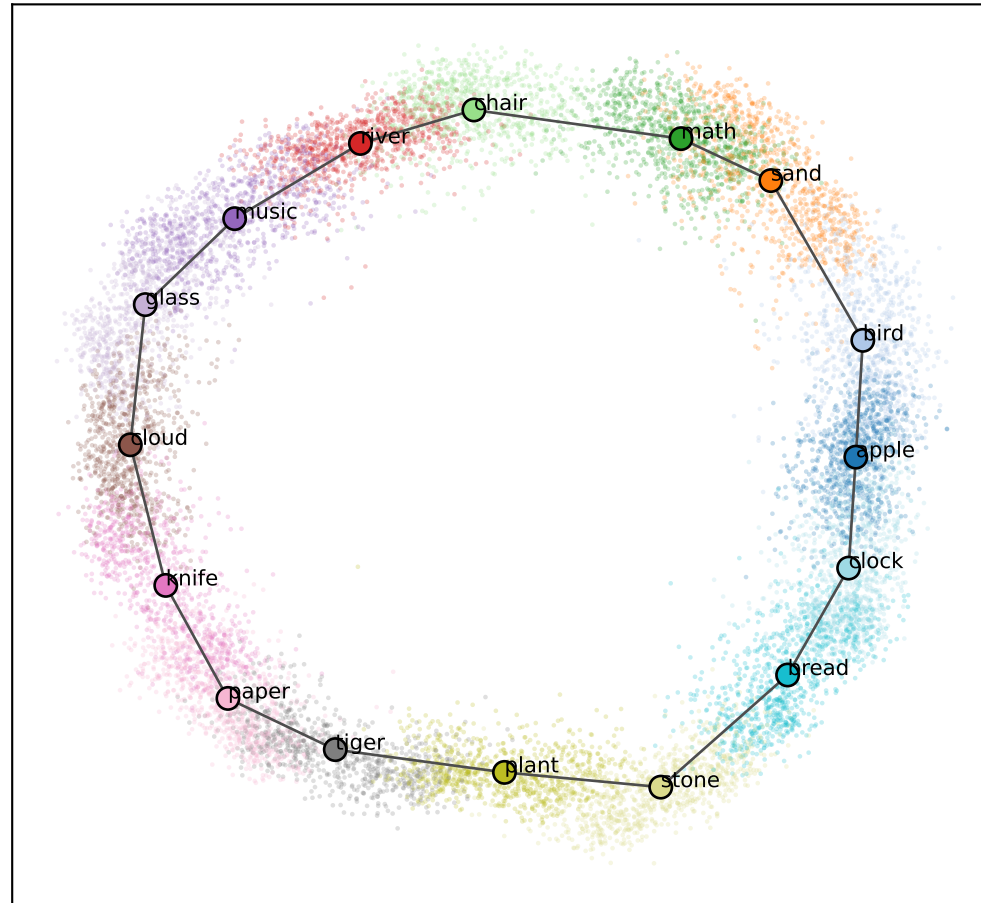


Gemma — every occurrence on the two best-2D axes (dots=occurrences, markers=node means)

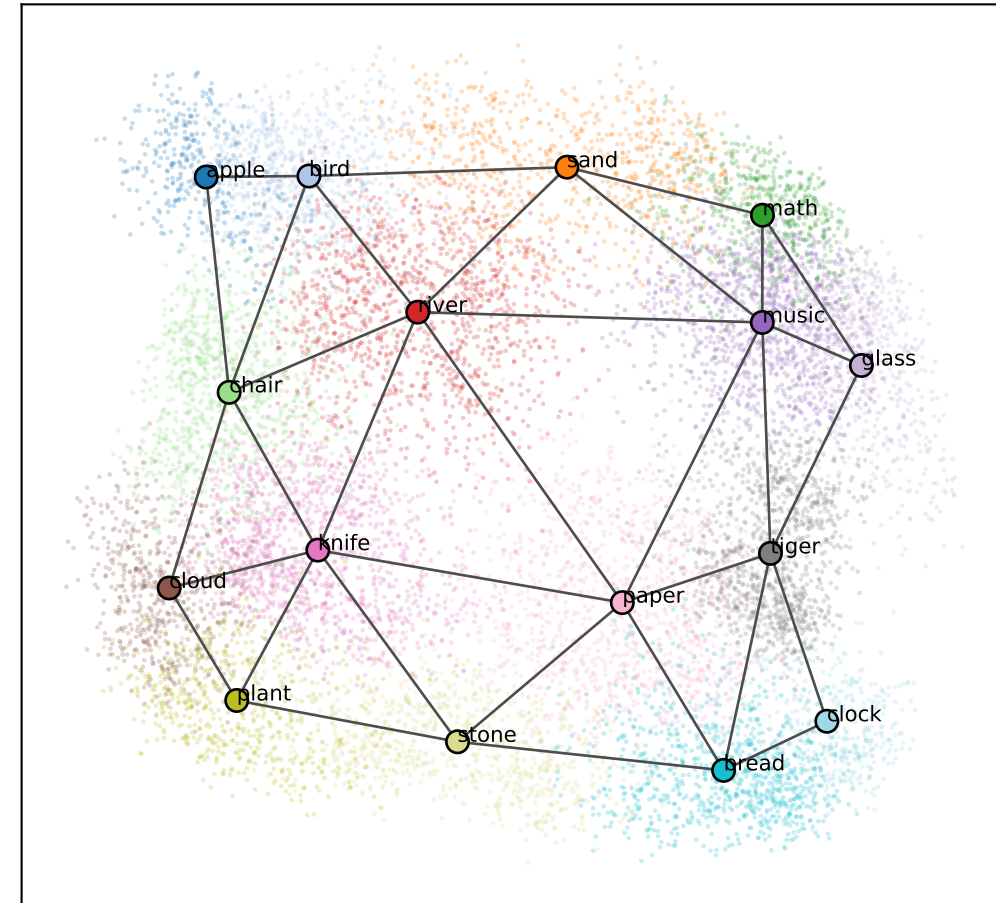
square_grid best-2D (RSA 0.88, $p < .001$)



ring best-2D (RSA 0.98, $p < .001$)

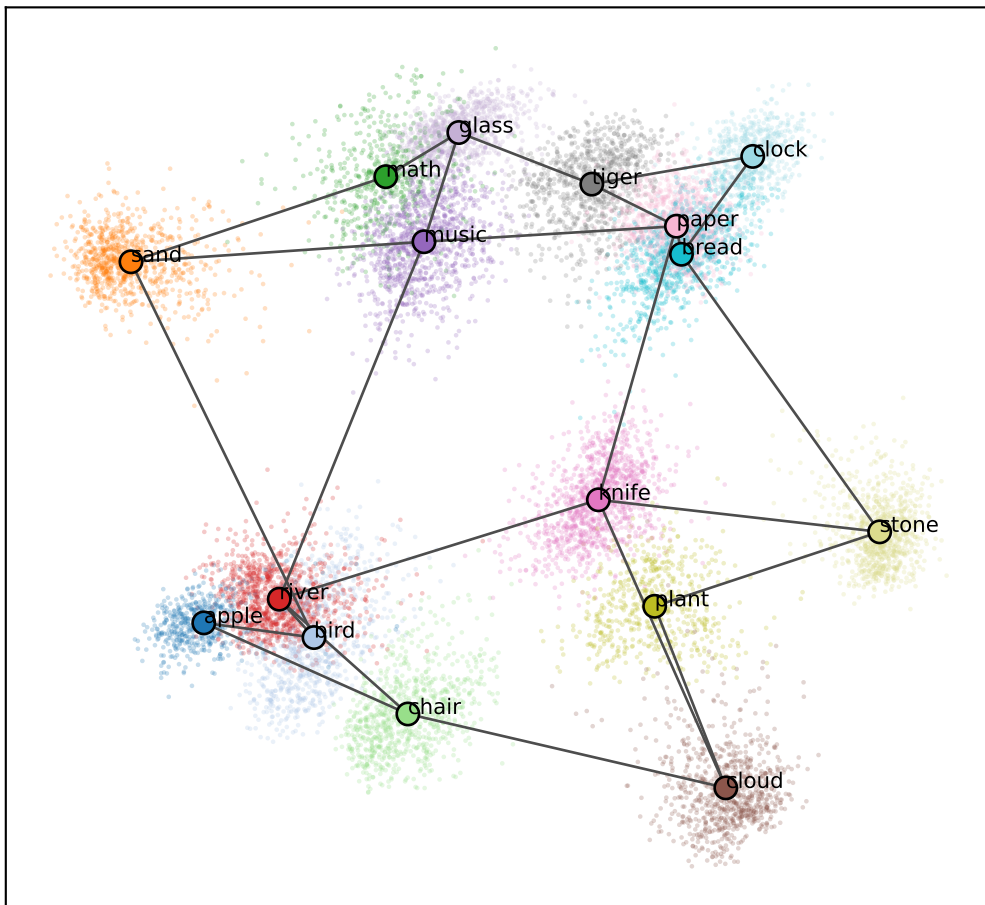


hex best-2D (RSA 0.90, $p < .001$)

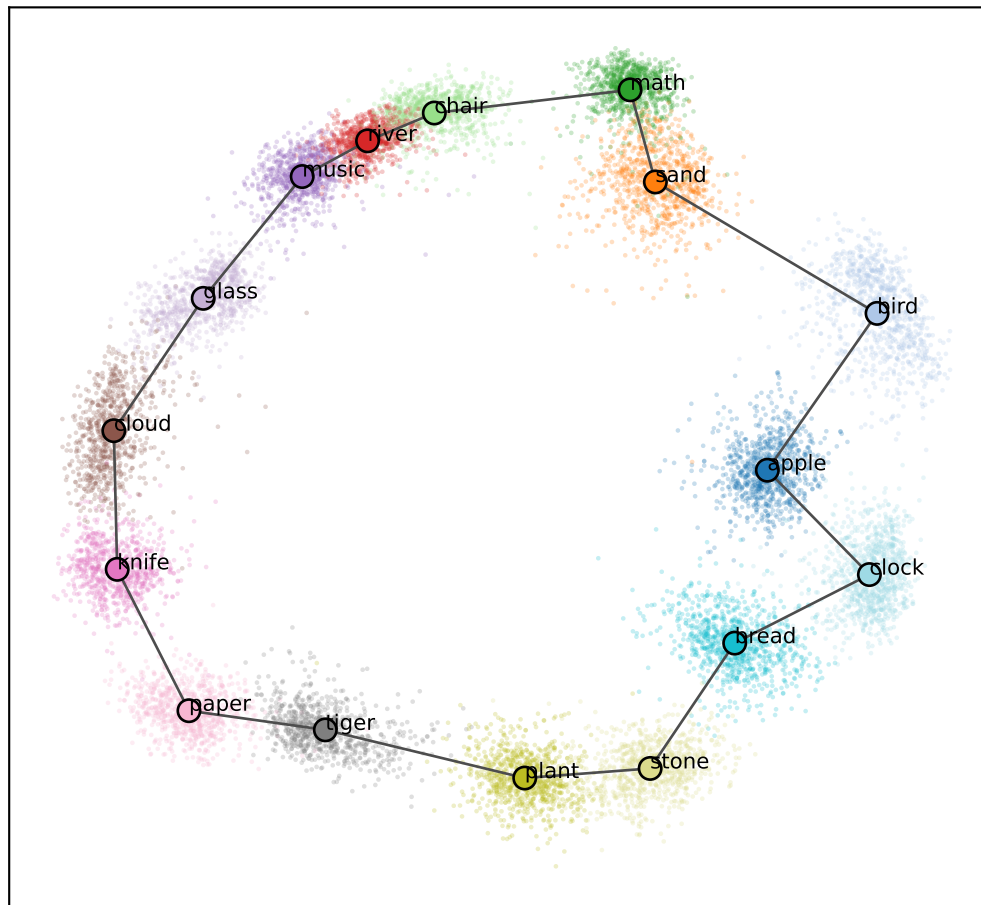


Qwen — every occurrence on the two best-2D axes (dots=occurrences, markers=node means)

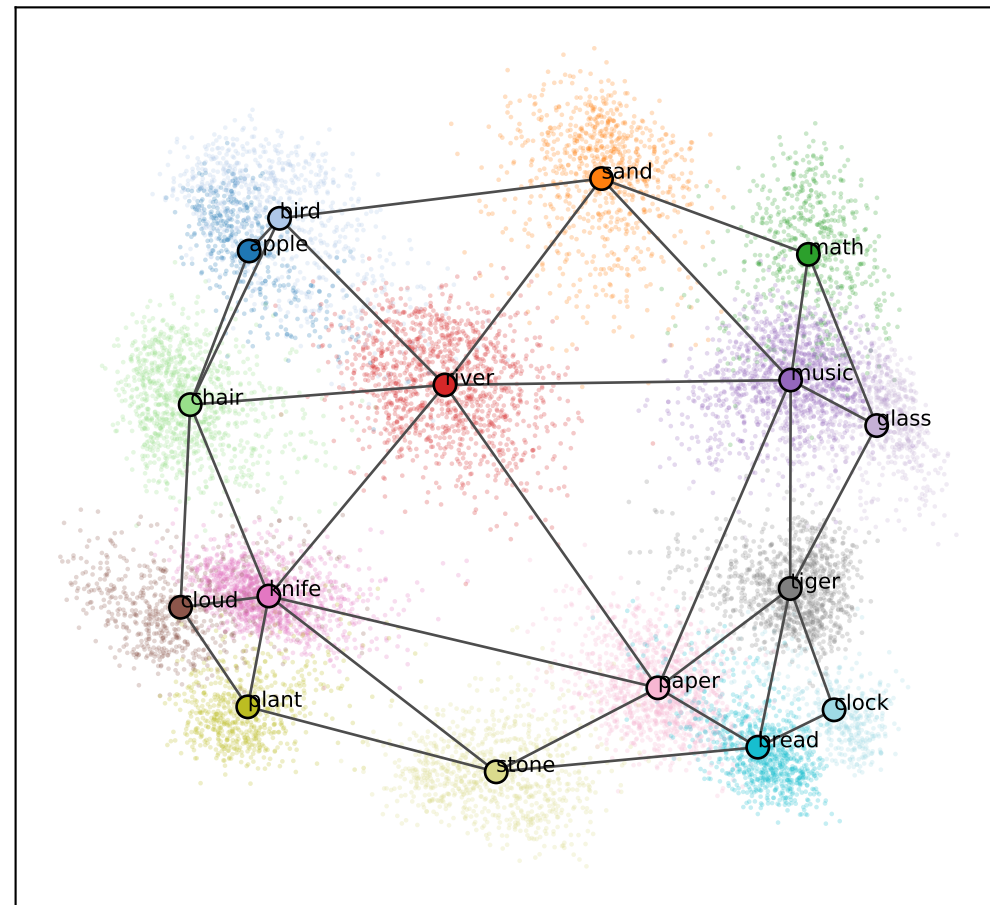
square_grid best-2D (RSA 0.72, $p < .001$)



ring best-2D (RSA 0.94, $p < .001$)



hex best-2D (RSA 0.85, $p < .001$)



best-2D RSA vs label-permutation null (gray = null $\leq$ 99th pct, red = null 95th pct)

